

A PUF based Authentication Technique for IoT Devices

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Abstract—The Internet of Things (IoT) have grown tremendously in last few decades due to the availability and low cost of Internet and sensors. The authentication of these large numbers of IoT devices is very important and has security concern since they are mainly deployed in open insecure environment. Also, the security technology that is being used by the existing servers cannot be applied to them since they operate with limited processor, memory and power and they only operate with lightweight or minimal security functions making them vulnerable for many security incidents also traditional authentication process mainly focuses on device to server authentication. Hence a novel device to device authentication scheme for IoT devices using Physical Unclonable Function (PUF) have been proposed which provides mutual authentication between IoT devices where authentication server acts as a reliable intermediary. Various security analysis has been discussed which can be defended by using proposed method.

Index Terms—Device to Device Authentication, Physical Unclonable function, Internet of Things.

I. INTRODUCTION

Internet of Things (IoT) is a network of large number of devices connected with each other for information exchange by using internet to have some value-added service [1]. In recent years, due to the widespread availability of Internet and low cost of sensors IoT devices are being used in almost every sector including smart home, smart cities, sensor-based weather monitoring, smart health monitoring etc. to provide convenience to human lifestyle.

However Increased demand of IoT have given rise to a new area of challenges and security. [2] The communication of IoT devices occurs over insecure public environment where unauthorized access can result in loss of confidentiality and lead to serious threat of overall network.

Because majority of the devices on IoT are not full-fledged computers rather these devices run on limited resources i.e., have limited processor speed, memory, power and capabilities. This is why the existing security technology used by computers and servers can't be applied to these devices. Instead, they are applied with lightweight security technology or minimal security functions or even no security functions at all. [3] As a result, various types of security attacks such as information leakage due to hacking and distributed denial of attacks occur in these vulnerable IoT nodes as well as these devices are vulnerable to various damages caused by

illegal copying and forgery of data leading to necessitate device-to-device authentication and secured transmission.

Moreover, one of the most important aspect for secure authentication is hardware trust of IoT devices. Each device has to have its unique identity to make these IoT device secure. Presently Physically Unclonable Function (PUF) is emerging as a valuable primitive that uses the hardware itself as the medium to generate an unique secret key, [4] thus making it impossible for one to clone the hardware that generates the exact same key [5]. PUF's basic principle comes from device level variations that occur during chip manufacturing process due to variation in channel doping concentration, variation in dielectric thickness, source/drain doping concentration, channel width, channel length etc. This kind of nano-scale variation is unpredictable and unique for each and every hardware. Its impossible to manufacture exactly the same PUF with similar differences even though it is made after processing through same wafer, same process, same die or same fab. [1] So for same input (Challenge) different PUFs are going to produce different output (Response) which acts as their fingerprint which can be used for authentication of the IoT devices [6].

In this paper, a new PUF based authentication scheme have been proposed which performs device to device authentication of IoT devices and generate session keys for secure communication. Our contributions in this paper are as follows :

- This paper proposed a PUF-based mutual authentication method between two IoT devices and session key is established for secure communication between these devices.
- The Authentication server is used as reliable intermediary which stores Challenge-Response pairs for each of the IoT devices.
- The proposed method uses arbiter PUF for generation of Challenge Response Pair.
- The use of error correction code (CRC) is proposed while generating response by PUF to make the response reliable and thus suitable for IoT devices even placed in various physical condition.
- The proposed protocol for IoT devices is resistant to physical threats.

The rest of the paper is organized as follows: Section II presents the related works, Section III describes the proposed authentication scheme, Section IV discusses security analysis and finally Section V is the conclusion of the paper.

II. RELATED WORK

Chatterjee et al. [7] proposed the PUF based key agreement protocol between device and server for secure communication of two IoT devices that uses a authentication server. During key agreement, public key was being generated for every device using PUF. However, their protocol is not resistant to impersonation, man in the middle, and replay attack [8].

Wallrabenstein [9] used PUF along with elliptic-curve cryptosystem on IoT device to generate private key when required to ensure tamper resistance rather than storing private key in the device memory. However, it needs some changes to be done in hardware of the IoT device.

The scheme proposed by Huth et al. [10] does not require much space for storage but it must involve significant hardware modifications for its use on the device.

A PUF based authentication model was proposed by Ashish et al. [11] for IoT devices for healthcare sector. In this scheme, the device plays the role of prover while the server serves as verifier. First a challenge is applied and out of it a set of sub-challenge is obtained through a Pseudo Random Generator. The system functions by verifying the device and the server. In their work, Yoon et al [3] presented an approach for mutual authentication between device to device for IoT environment. This involves using only one initial challenge response pair stored at the server for each device, updated later with secure communication from device but as changing circumstances often alters PUF responses and necessitates stability, an error correction mechanism is required for stable response which was not included in their proposed work due to which a reliable response may not be updated in server creating authentication issues.

Another approach that has been used in this area [12] is the fuzzy identity based Encryption(IBE), where it exploits the channel state information to create a unique, fuzzy identity for devices IoT devices. however handling the fuzziness of physical-layer information creates inconsistency in the extraction of identity unless carefully dealt with.

Another research by Hu et al [13] focuses on anonymous authentication based on Elliptic Curve Cryptography. However it is vulnerable to side-channel attack and also it is difficult to scale to very large IoT networks.

Many of the existing PUF based IoT device authentication scheme have been studied focused on authentication between devices and servers. However in various IoT Scenarios while one device needs to communicate with other device, it becomes more essential that device should mutually authenticate each other (Device to Device Authentication) and securely communicate with each other rather only authentication between device and server (Device-to-Server Authentication).

III. PROPOSED SCHEME

This paper presents a PUF based mutual authentication method between two IoT devices. Instead of performing authentication only between device and server, the device to device authentication is performed using authentication server as a trusted intermediary. The authentication scheme consists of two phases Enrollment phase where device ID, challenge and response is stored in authentication server in a secure environment. Next phase is authentication and key exchange phase where both the devices mutually authenticate each other with the help of authentication server as intermediary and both devices shares secure session key which is used to encrypt communication messages between the IoT devices in order to defend various attacks.

PUF Design: Uniqueness and reliability are the two key quality metrics for PUF designs. For design criteria and implementation, inter-chip and intra-chip distances were used for uniqueness and reliability. Inter-chip Hamming distance D_{inter} is a metric for determining difference between different manufactured PUF instances hence reveal the usage of static variation that occurs as a result of the manufacturing process. In an ideal condition, the average difference will be around 50% for all the physical PUF instances because anything above or below that ratio would indicate some form of bias in response. As shown in Equation (1) by using same set of challenges, it is computed as an average hamming distance across different PUF instantiations. Here, C represents common challenge used during uniqueness evaluation and the number of PUF instances is represented by m .

$$D_{inter} = \frac{2}{m(m-1)} \sum_{i=1}^{m-1} \sum_{k=i+1}^m D(r_i, r_k) \quad (1)$$

where $r_i = PUF_k(C)$, $r_k = PUF_k(C)$

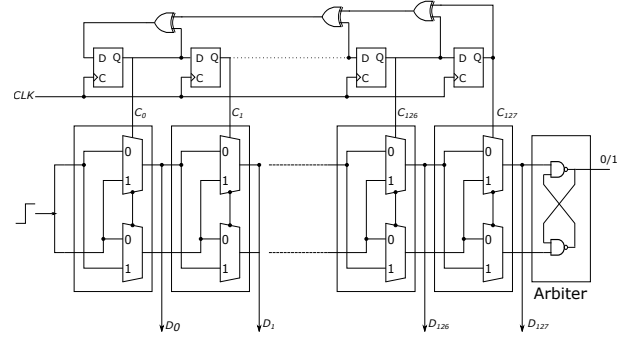


Fig. 1. Structure of a n bits Arbiter PUF

The use of Cyclic Redundancy Code (CRC) removes the error / noise in PUF response.

Since the introduction of the first silicon PUF, known as Arbiter-PUF (APUF) in 2002, the PUF community has been working on what are called strong PUFs. This is probably the most researched silicon PUF implementation and has been looked at and built from the early days of PUF research. These have a large challenge-response pair (CRP) space and

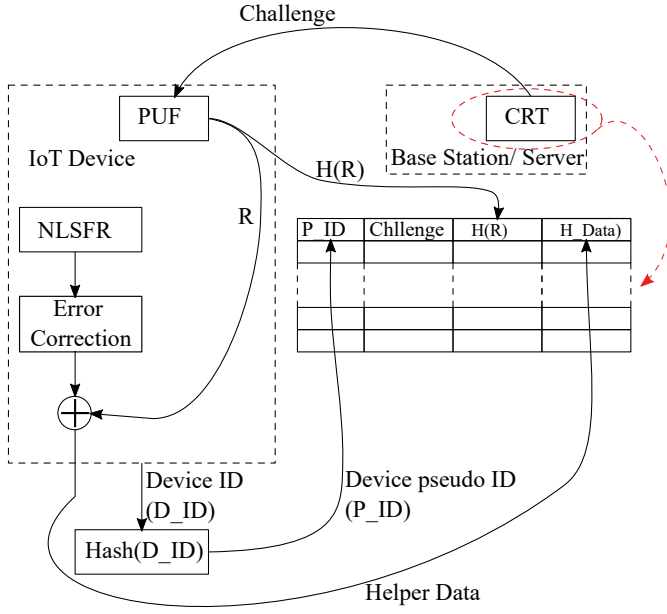


Fig. 2. Structure of Challenge-Response Table (CRT) for PUF-based device enrollment phase.

resistant to modeling attacks. One of the most researched and implemented silicon PUF designs is the Arbiter PUF (APUF) that has been used extensively in PUF studies. APUFs offer advantages such as minimum hardware overhead, ease of design and regularity making it more attractive for VLSI implementation. The most common compositions are the XORPUF, Light Weight Secure PUF (LSPUF), Multiplexer PUF (MPUF), as well as the latest newly suggested Interpose PUF (IPUF). Thus, APUF can be used as a building block to form more secure APUF composition. A single APUF consists of a sequence of different two port delay stages that are all structurally identical with an arbiter (usually in form of latch) at its end stage. Hybrid PUF-finite state machine (PUF-FSM) is used to build controlled strong PUF. This structure eliminates the need for error correction logic and corresponding computation, the need to hold helper data in memory, and loading it on-chip responses that are error-free which are computed without an APUF having a larger challenge response pair space.

A. Network Model

The network model for PUF based device to device mutual authentication between IoT devices consists of an authentication server, which act as powerful, secured intermediary without any resource constraint. It stores Challenge Response Table (CRT) securely for each IoT devices using hashing technique. It is connected to various resource constraint IoT devices which contains embedded arbiter PUF inside them using which PUF response can be generated for a given challenge. These IoT devices can communicate with each other and also with authorized server through internet using suitable routers.

TABLE I
CHALLENGE RESPONSE TABLE (CRT) STORED IN SERVER

Hash(ID)	Challenge	Hash (Response)
$H(ID_A)$	C_{A1}	$H(R_{A1})$
$H(ID_A)$	C_{A2}	$H(R_{A2})$
...	...	...
$H(ID_B)$	C_{B1}	$H(R_{B1})$
$H(ID_B)$	C_{B2}	$H(R_{B2})$
...	...	...

B. Enrollment Phase

In the Enrollment phase, before deploying IoT nodes, in a secure environment each IoT device shares challenge and response pair with authentication server which is considered as a authenticated and trusted party.

The authentication server stores the Hashed ID of each device along with its challenge and Hashed Response in the Challenge Response Table (CRT) as shown in TABLE I. A random challenge is being selected from CRT for two communicating devices whenever request for authentication is received.

C. Authentication & Key Exchange

This is performed in the second phase when all the IoT nodes have been deployed. Now when a IoT device (suppose device A) wants to communicate with other IoT device (suppose device B), it will perform the following steps for performing mutual authentication and generating secure session key:

- 1) First Device A will send hashed value of its ID $H(ID_A)$ and a random number N_A generated by using NLSFR to device B.
- 2) Device B will forward Device A details along with its own ID $H(ID_B)$ and a newly generated random number N_B to authentication server.
- 3) Authentication Server will search for hashed value of ID for both device and selects a random challenge-hashed response pair $(C_{A1}, H(R_{A1}))$ and $(C_{B1}, H(R_{B1}))$ for each of them and generates secure key K_{A1} and K_{B1} using concatenated challenge & hashed response pair which is secret to each device, generates a random number N_S and responds encrypted N_S concatenated with N_A and N_B separately along with challenges used C_{A1} and C_{B1} to device B.
- 4) Device B upon receiving response message from server, uses challenge C_{B1} to generate response through embedded Arbiter PUF. The response is passed through Cyclic Redundancy Check (CRC) for error correction and stable response R_{B1} is generated. By concatenating hashed value of response $H(R_{B1})$ and Challenge C_{B1} secret key K_{B1} generated which is used to decrypt response and extract and compare N_B if matched Server is authenticated.
- 5) After authenticating server, Device B generates a symmetric key $K_S = H(N_A || N_S)$ and forwards received

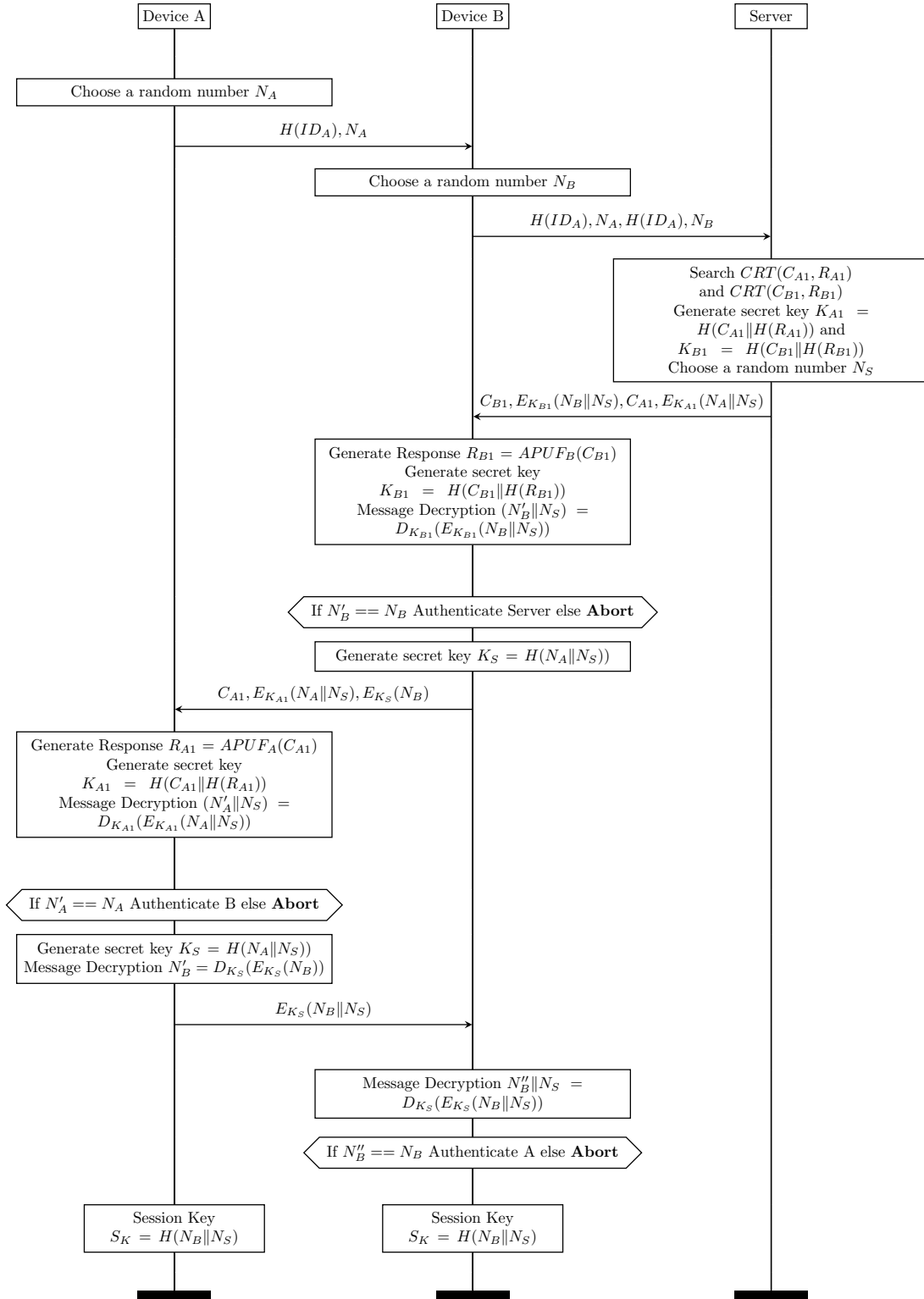


Fig. 3. Details of the Device Authentication

message from server C_{A1} , encrypted $(N_A||N_S)$ with key K_{A1} along with N_B encrypted with K_S to device A.

- 6) Device A again performs similar steps as done by device B, generates response R_{A1} using C_{A1} received using APUF and CRC, generates key K_{A1} using $H(C_{A1}||H(R_{A1}))$ and decrypt response received to extract and compare N_A if matched Device B is authenticated.
- 7) After authenticating Device B, Device A generates the symmetric key $K_S = H(N_A||N_S)$ and decrypt and extract N_B which will be used for generate session key for secure communication with device B. Device A encrypts $(N_B||N_S)$ using symmetric key K_S and send to Device B.
- 8) Device B decrypts the recieved response from device A using symmetric key K_S and extracts and compare N_B if matched device A also authenticated.
- 9) After both the devices are mutually authenticated. Both device uses mutually shared N_B and N_S to generates secure session key $SK = H(N_B||N_S)$. Now all the communication between these devices will be encrypted by the session key SK to avoid any attack.

PUF based Authentication: The PUF-based authentication between the IoT device and server provides high-level device security without storing PUF challenge-response. Traditional PUF-based authentication techniques have drawbacks since they only function with challenge-response pairs (CRPs) disclosed to the verifier, breaching the ideal of anonymity.

The novelty of this lightweight PUF is to build a secure communication model for medical IoT devices, which are very much vulnerable in wireless communication. This model will help to protect from unauthorized access, and protect the device identity and device authentication.

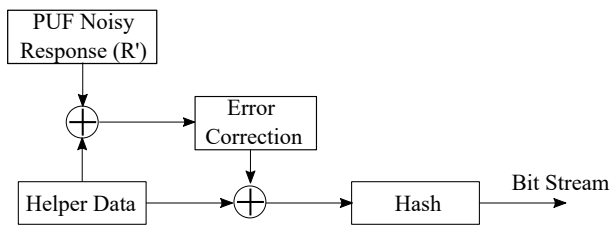


Fig. 4. Structure of Challenge-Response Table (CRT) for PUF-based device Authentication phase.

IV. SECURITY ANALYSIS

The proposed authentication scheme is robust and secured. This scheme prevents various attacks possible on IoT devices as described below:

A. Resistance against information leakage

By using PUF, threat of Information Leakage have been prevented. In conventional authentication secret key need to be stored separately which can be compromised leading to information leakage but since PUF generates unique hardware

based response which can be used as fingerprint to authenticate a device. Moreover hashed value of responses for each challenge is stored so that even if the server is compromised the secret response is not exposed minimizing information leakage.

B. Violation of Perfect Forward Security

In order to implement Perfect Forward Security static challenge response pairs of PUF has not been used for generating session keys instead random numbers generated through server and devices are hashed and used as session keys. For every session a new session key is generated. This would ensure that session keys are derived from temporary keys that are not stored long-term, and even if the PUF responses are compromised, past session keys would remain secure.

C. Resistance against modelling attack

This scheme prevent from modelling attack [14] since authentication process does not expose Challenge-Response Pair (CRP) directly instead secret key based on CRP is used for authentication also in authentication server instead of plain response hashed value of response is kept blocking any machine learning based modelling attack.

D. Resistance against replay attacks

This scheme also prevents replay attacks, since mutual authentication is performed between IoT devices for every session before communication.

E. Resistance against man in the middle attack

Since communication between devices is encrypted using session key which cannot be decrypted without knowing the hardware generated CRP for each device for each session, which is very difficult. So this authentication scheme is safe against man in the middle attack, spoofing attack and impersonation attack.

F. Resistance against cloning attack

This authentication method prevents from cloning attack [15] (physical attack) since each IoT devices is embedded with PUF module, cloning of which is not possible thus preventing from cloning attack.

G. Resistance against side channel attack

PUF is a hardware made up of circuits so it may be prone to side channel attack but if the PUF chip embedded in IoT device is equipped with countermeasures against side channel attack [16], this scheme will be able to prevent any hardware based side channel attacks.

H. Resistance against Empirical Security key Leakage

Conventional authentication schemes often store sensitive keys in non volatile memory making them vulnerable by extraction attack. The PUF-based scheme decreases dependency on non-volatile memory by employing PUF that generates its responses dynamically, hence preventing empirical key leakage.

I. Resistance against Traceability attack

Traceability attacks occur when an adversary can track a device by observing its authentication messages over time. In order to mitigate the risk of traceability attacks, in proposed scheme the server generates random challenges for each authentication session. Since these challenges are unpredictable, the corresponding responses from the device will also be unpredictable, making it difficult for an attacker to establish a pattern or link sessions.

J. Resistance against DDoS attack

Since PUF generates unique responses for each IoT devices based on physical characteristics. This uniqueness makes it difficult for attackers to spoof identity and launch large scale attacks using compromised fake identities commonly used in DDos attack.

K. Resistance against Stolen Device attack

PUF-based scheme does not require storing static secret keys in device memory. Hence, if a device is stolen, the risk of key extraction is reduced because authentication depends on dynamic PUF responses. Also since the challenges are random and responses are unique, simply having the device does not guarantee successful authentication without access to the correct PUF response mechanism.

L. Resistance against Privilege Intruder attack

This attack involves attacker gaining unauthorized access to system which attempts to elevate their privileges to misuse the system. Since the core of the proposed scheme is PUF which generates unique response based on physical properties of PUF which are highly unpredictable and difficult to replicate it becomes difficult for the intruders from forging valid authentication responses. Additionally scheme does not rely on storing static secret keys hence it reduces the risk of key extraction and misuse by an intruder who has gained access to the device or system.

M. Resistance against Impersonation attack

To prevent an adversary attempting to masquerade as a legitimate device to gain unauthorized access, a unique PUF is associated with each IoT device such that responses to specific challenges are unique and cannot be replicated by another device or attacker.

N. Resistance against Privacy Theft

It unauthorized access to sensitive information. In proposed scheme instead of storing static keys or passwords, the device uses its PUF to generate responses on-the-fly. This reduces the risk of privacy theft through physical attacks on stored data also session keys are used to secure device to device communication.

V. CONCLUSION

In this paper, we have seen the proposed device to device authentication model for IoT devices using PUF. The proposed model does unique identification of IoT nodes using arbiter PUF. The hashed value of response corresponding to challenge stored in server along with hashed device ID thus thwarting any modelling attack, also a session key is established for creating secure channel for communication between IoT devices thus secured encrypted message is being communicated after proper authentication defending various possible security attacks.

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